

SAMANTHA FERNANDEZ-PEREZ

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EXPERIENCE

Research Intern, Dept. of Epidemiology Weill Cornell Medicine, NY Jun – Aug 2026

- Preprocess high-dimensional multi-omics TCGA-OV data and engineer women-specific clinical features (BRCA1/2, HRD, platinum response) to fine-tune the SeNMo deep learning model for ovarian cancer survival prediction
- Implement and evaluate statistical and machine learning models in Python and PyTorch using stratified cross-validation, benchmarking against the published 0.518 C-index baseline
- Generate Kaplan-Meier curves and HGSOc subgroup analyses to interpret model performance and prepare research outputs for publication

Research Assistant, Dept. of Biochemistry & Molecular Biology Syracuse, NY Sep 2024 – Oct 2025

- Built a high-throughput screening system for prokaryotic peptide and protein libraries against SUMO and ULP therapeutic targets at Upstate Medical University SUNY
- Optimized protein purification and molecular biology protocols, validating binding candidates to support scalable, cost-effective therapeutic development

HB Experimental Forest Research Assistant Woodstock, NH Apr – Aug 2024

- Analyzed seasonal and elevational variations in particulate organic carbon and subsurface nanoparticles, documenting wollastonite's (CaSiO₃) role in mitigating acid rain
- Led TACNY STEM Trekkers Summer Camp sessions on lake stratification and seasonal water chemistry

Pediatrics Administrative Assistant Queens, NY Aug 2018 – Jun 2021

- Supported clinical staff with sterile room preparation and patient care logistics across pediatric appointments
- Translated between English and Spanish for patient families, enabling clear communication during care decisions

EDUCATION

Syracuse University, College of Engineering and Computer Science

Bachelor of Science in Biomedical Engineering

Coursework: Bioengineering Fundamentals*, Engineering Computational Tools*, Biomechanics, Medical Devices & Biomaterials, Quantitative Physiology, Biomedical Systems, Signal & Control*, Bioinstrumentation, Electrical Engineering*, Global Regulatory Affairs (* with Lab)

Extracurriculars: Syracuse University Waterpolo and Swim Club, Boxing Club, iGEM (International Genetically Engineered Machine), Women in Science and Engineering (WiSE), Society of Hispanic Professional Engineers (SHPE), Biomedical Engineering Society (BMES)

SKILLS

Software: Python, MATLAB, Jupyter, Anaconda, R-Studio, Fusion360, Canva, Microsoft Word, PowerPoint, Excel

Biolaboratory: Electroporation, DNA Gel Electrophoresis, Genetic Therapy, Heat Shock Transformation, Cell Culture

Environmental Lab: Ion Chromatography, Dynamic Light Scattering, Nitrogen and Carbon Analysis

Languages: Native Spanish, Fluent English, Conversational French and Portuguese

Certifications: CITI Health Information Privacy and Security for Clinical Investigators (HIPS)

ENGINEERING APPLICATIONS

Prosthetic Sixth Finger Syracuse University Jan – Feb 2024

- Designed in Fusion360 and programmed an Arduino-controlled prosthetic finger with temperature sensing and real-time patient feedback

Coding Operating Python Syracuse University Jan – May 2023

- Built and tested a Python and microprocessor-based assistive device with two teammates, enabling computer control without fine motor input

LEADERSHIP

iGEM President and CEO Syracuse University Jan 2023 – Present

- Founded Syracuse University's first iGEM team and recruited an interdisciplinary student group
- Led proposal development for international competition across multiple academic departments